Equilibrium

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1 Reaction Rates

Atoms, ions, and molecules react with one another via collisions. The more frequent and energetic the collisions, the faster the reaction rate.

Take the formation of compound C as an example:

$$A + B \longrightarrow C$$

The rate at which C forms depends on factors such as concentration of A, concentration of B, temperature, the presence of a catalyst, etc. and can be expressed mathematically as:

$$\text{Rate} = k[A]^m[B]^n$$

Here, k is the rate constant, which is specific to the reaction and depends on temperature and the presence of a catalyst (and other factors if applicable). And m and n are the reaction orders, which represents the magnitude of effect that the given reactant has on the reaction rate. Here is how a hypothetical graph of the concentration of A, B, and C over time would look like:

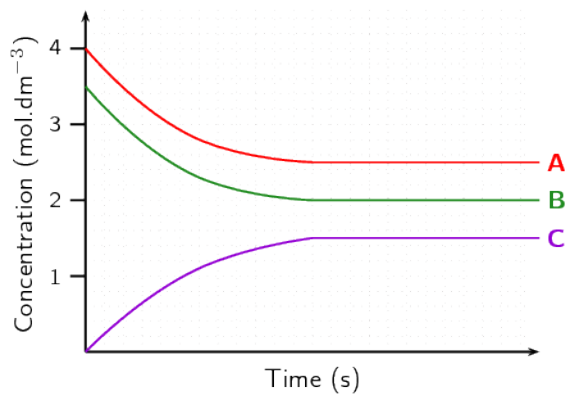

Figure 1: ABC

Definition 1.1: placeholder

Generally in elementary reactions (single-step reactions), the **reaction orders (m and n) are equal to the stoichiometric coefficients** of the reactants in the balanced chemical equation. However, in complex reactions (multi-step reactions), although not covered intensely in this context within this course, the reaction orders may not directly correspond to the stoichiometric coefficients and must be determined experimentally.

Let's take the real world example of the decomposition of H_2O_2 ($H_2O_2 \longrightarrow H_2O + O_2$) as an example. When you graph the concentration of H_2O_2 over time, you get an exponential decay curve.

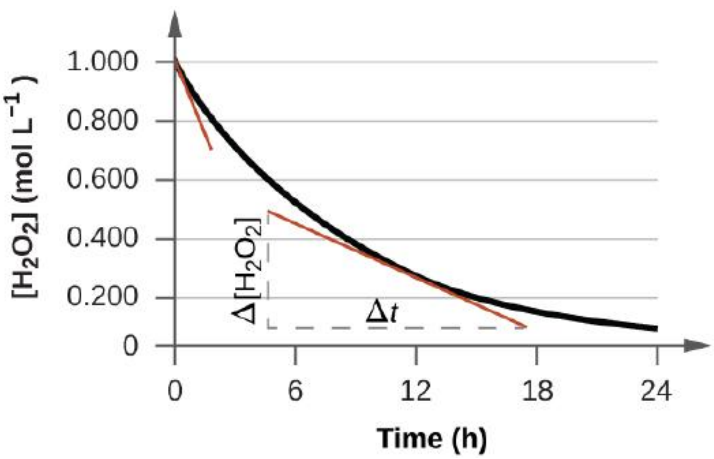

Figure 2: H2O2 decomposition

As you can see, the rate at which H_2O_2 decomposes slows down over time as $[H_2O_2]$ decreases.

2 Chemical Equilibrium

Reactions do not always proceed in only one direction. Many reactions are **reversible**, meaning that the products can react to reform the reactants. When the rates of the forward and reverse reactions are equal, the system is said to be at **equilibrium**.

Thus, for the general reversible reaction $AB \rightleftharpoons A + B$, the equilibrium condition can be expressed as:

$$k_f[\mathrm{AB}] = k_b[\mathrm{A}][\mathrm{B}]$$

The equilibrium constant, K_c, is defined as:

$$K_c = \frac{k_f}{k_b} = \frac{[\mathrm{A}][\mathrm{B}]}{[\mathrm{AB}]}$$

K_c provides insight into the position of equilibrium. A large K_c ($>> 1$) indicates that the reaction favors the formation of products, while a small K_c ($<< 1$) indicates that the reaction favors the reactants.

Some examples of reactions that possess a **large** K_c (reaction to completion) include:

$Si + O_2 \longrightarrow SiO_2 \quad K_c \approx 10^{150}$

$2\,H_2 + O_2 \longrightarrow H_2O \quad K_c \approx 10^{81}$

Some examples of reactions that possess a **moderate** K_c (half-reaction) include:

$CO + \text{Hemoglobin} \longrightarrow CO \cdot \text{Hemoglobin} \quad K_c \approx 300$

$O_2 + \text{Hemoglobin} \longrightarrow O_2 \cdot \text{Hemoglobin} \quad K_c \approx 4$

$N_2O_4 \longrightarrow 2\,NO_2 \quad K_c \approx 0.2$

An example of a reaction that possess a **neglible** K_c (no reaction) include:

$N_2 + O_2 \longrightarrow 2\,NO \quad K_c \approx 10^{-31}$

Let's consider the decomposition of Bromothymol Blue (which will be denoted as A) as an example:

$$HA \rightleftharpoons H^+ + A^-$$

Visually, HA is yellow, while A^- is blue. If we start with a solution that is purely HA, as the reaction proceeds towards Eq, the solution will gradually change color from yellow to green to blue as more A^- is formed.

If hypothetically, the Eq constant for this reaction is $K_c = 4$, this means that at Eq, the concentration of A^- will be four times that of HA. Thus, the solution will appear predominantly blue at Eq. Conversely, if $K_c = 0.25$, the concentration of HA will be four times that of A^-, and the solution will appear predominantly yellow at Eq.

3 Le Chatelier's Principle

Le Chatelier's Principle states that a system in Eq subjected to a stress will react in a direction that tends to reduce that stress. Stresses can include changes in **concentration**, **pressure**, **volume**, or **temperature**.

First, consider the effect of changing **CONCENTRATION**. Let's reexamine the Bromothymol Blue example:

$$HA \rightleftharpoons H^+ + A^-$$

If we add more H^+ to an initially green solution (meaning equal parts HA and A^-), the system will shift left to consume the excess H^+, resulting in a more yellow solution. Conversely, if we remove H^+ (by adding OH^-), the system will shift right to produce more H^+, resulting in a more blue solution.

Consider another example ($PCl_5 \longrightarrow PCl_3 + Cl_2$) based on the visual below:

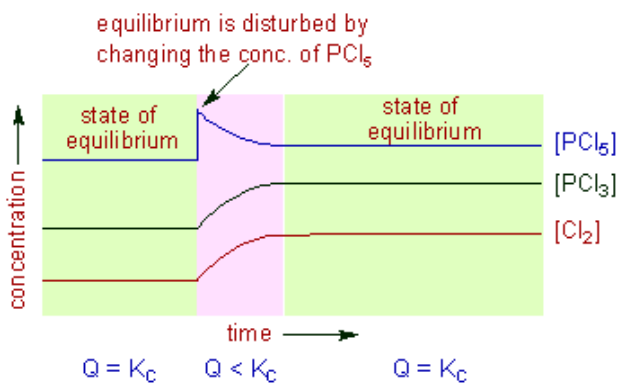

Figure 3: lc conc

As the abrupt addition of PCl_5 decreases the reaction quotient (Q) below K_c, the system shifts right to re-establish Eq, resulting in a subsequent increase in the concentrations of PCl_3 and Cl_2.

Second, consider the effect of changing **PRESSURE** or **VOLUME**. For gaseous reactions, increasing pressure (by decreasing volume) will shift the Eq towards the side with fewer moles of gas, while decreasing pressure (by increasing volume) will shift the Eq towards the side with more moles of gas.

Definition 3.1: Example

The K_p (equilibrium constant in terms of partial pressures) is how we express the Eq constant for gaseous reactions. Instead of using concentration, we use partial pressure.

Consider the following reaction:

$2\,NO_2(g) \rightleftharpoons N_2O_2(g) \qquad K_p \approx 7$ at 25°C

$K_p = \frac{P_{N2O2}}{(P_{NO2})^2}$

If we increase V, P decreases. The system will consequently shift left to produce more NO_2 to counteract the decrease in P by increasing the total moles of gas in the system. This is also represented mathematically as the reaction order for NO_2 being 2, meaning that changes in its concentration (or partial pressure) have a squared effect on the reaction quotient (Q) whereas N_2O_2 only has a linear effect.

Lastly, consider the effect of changing **TEMPERATURE**. The direction in which the Eq shifts depends on whether the reaction is **exothermic** (*heat is a product*) or **endothermic** (*heat is a reactant*).

Consider the following exothermic reaction:

$2\,NO_2(g) \rightleftharpoons N_2O_4(g) + \text{heat}$

Just like adding a N2O4 would shift the Eq left, adding heat (increasing T) would also shift the Eq left to consume the excess heat. Conversely, removing heat (decreasing T) would shift the Eq right to produce more heat.

Endo- and exothermic reactions are covered more in depth in the Thermodynamics chapter.

4 Types of Equilibria

There are two main types of equilibria we will cover: homogenous and heterogeneous equilibria.

In **homogeneous equilibria**, all reactants and products are in the same phase (solid, liquid, or gas). Here are some examples.

$Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons FeSCN^{2+}(aq)$ Solution

$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$ Gas

$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$ Acid-Base

In **heterogeneous equilibria**, the reactants and products are in different phases. Here are some examples.

$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$ Solubility of solids

$O_2(g) \rightleftharpoons O_2(aq)$ Gas solubility

$H_2O(s) \rightleftharpoons H_2O(l)$ Phase changes

Here, we will take a closer look at **solubility equilibria** and **phase change equilibria**.

First, let's consider the solubility equilibria of **SOLID COMPOUNDS IN LIQUIDS**.

Consider the solubility equilibrium of $C_6H_{12}O_6$ (glucose) in water and its dissociation and crystallization rate equations. We call solutes that do not dissociate in solvent as **molecular or nonelectrolyte solutes**.

$C_6H_{12}O_6(s) \rightleftharpoons C_6H_{12}O_6(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[C_6H_{12}O_6(aq)]$

As you can see, the dissociation rate only depends on the surface area (SA) of the solid glucose, while the crystallization rate depends on both the SA and the concentration of dissolved glucose in the solution. When we reach equilibrium, these two rates are equal, thus

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [C_6H_{12}O_6(aq)]$$

For glucose, the K_{sp} value is 5.1M at 25°C. This means that at equilibrium, the maximum concentration of dissolved glucose in water is 5.1M. Beyond this concentration, any additional glucose will remain undissolved.

A different compound like NaCl works in the same way. We call these types of solutes that dissociate into ions in solvent as **ionic or electrolyte solutes**.

$NaCl(s) \rightleftharpoons Na^+(aq) + Cl^-(aq)$

Dissociation Rate = $k_f(SA)$

Crystallization Rate = $k_c(SA)[Na^+(aq)][Cl^-(aq)]$

At equilibrium, these two rates are equal, thus

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = [Na^+(aq)][Cl^-(aq)]$$

At 25°C, the K_{sp} value for NaCl is approximately 37.3. With a little algebra, we can determine that the maximum concentration of dissolved NaCl in water at equilibrium is approximately $6.11M$.

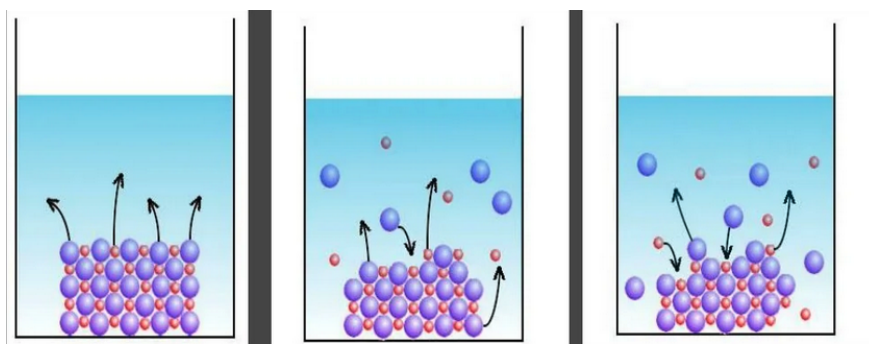

Figure 4: solid eq

Definition 4.1: Example

SA being the only factor in the dissociation and crystallization rate is actually a **simplification**. In reality, many other factors can also affect these rates. However, for the purpose of understanding solubility equilibria, focusing on SA provides a clear and straightforward model for the system.

Now, let's consider the solubility equilibria of **GASEOUS COMPOUNDS IN LIQUIDS**.

Consider the solubility equilibrium of CO_2 in water.

$CO_2(g) \rightleftharpoons CO_2(aq)$

When we reach equilibrium, we have the following:

$$K_H = \frac{[CO_2(aq)]}{P_{CO2}} \rightarrow [CO_2(g)] = K_H P_{CO2}$$

This equation is known as the **Henry's Law**. The constant K_H is called the **Henry's Constant** (M/atm). It is a number that is used to describe the solubility of a gas in a liquid. The higher the K_H value, the easier it is for a gas to dissolve in a liquid.

Henry's Law states that at Eq, the concentration of a gas dissolved in a liquid is proportional to its partial pressure in the gas phase.

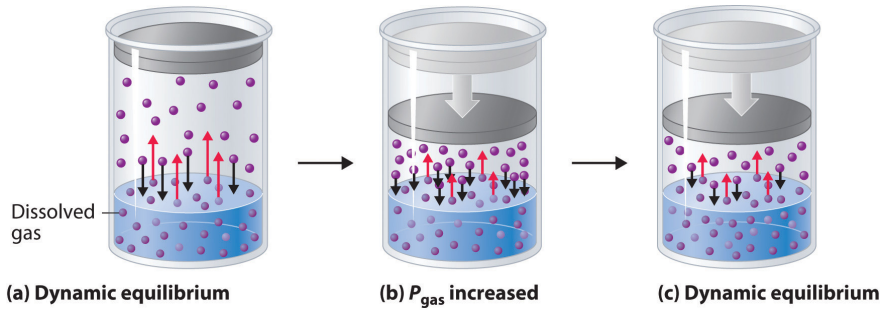

Figure 5: gas eq

Okay, now let's talk about **PHASE CHANGE EQUILIBRIA**. Consider the vaporization of water in a sealed flask.

$H_2O(l) \rightleftharpoons H_2O(g)$

Evaporation Rate = $k_f(SA)$

Condensation Rate = $k_c(SA)\ P_{H2O}$

Just as the solubility equilibria for nonelectrolyte solutes is only dependent on the solvent concentration, the phase change equilibria for a gas to liquid transition is only dependent on the water vapor pressure within a sealed container.

$$K_{sp} = \frac{k_c(SA)}{k_f(SA)} = P_{H2O}$$

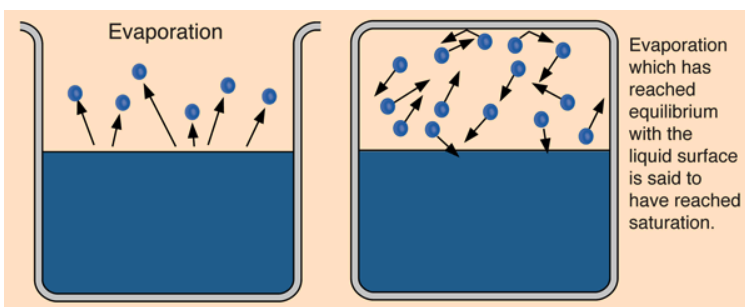

Figure 6: phase eq

Phase change equilibria can be explained in this matter only for the gas to liquid state transition.

5 Temperature Effects

You may have noticed that all the previously covered Eq scenarios took place at a constant T (25°C). However, what if the temperature of the system changed?

Just like a chemical compound is either a reactant or product within a chemical equation, heat behaves the same way. Therefore, we can think of an increase in T as the addition of heat to the system, and a decrease in T as the removal of heat from the system.

When heat is released from a reaction (heat is on the product side), the reaction is called **exothermic**. When heat is consumed from a reaction (heat is on the reactant side), the reaction is called **endothermic**.

In the context **phase change equilibria** (across solid, liquid, and gaseous transitions), an increase in T fundamentally increases the KE of the compound. Under constant P, phase changes represent a point of equilibrium where the strength of the KE of the molecules intersect with the strength of the IMFs holding them together.

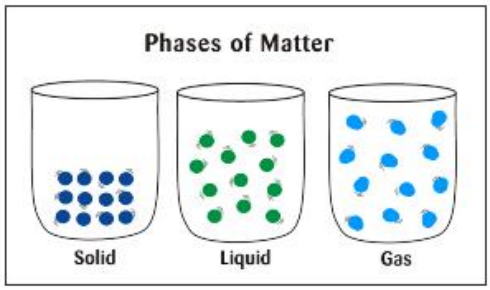

Figure 7: phases

When we add a dynamically changing P to the system, the resulting temperatures at which phase changes occur also change. Take water for example. The phase change temperatures we are all familiar with are 0°C and 100°C. This is only the case when P remains constant at 1 atm. If we shift P, the phase change temperatures change accordingly.

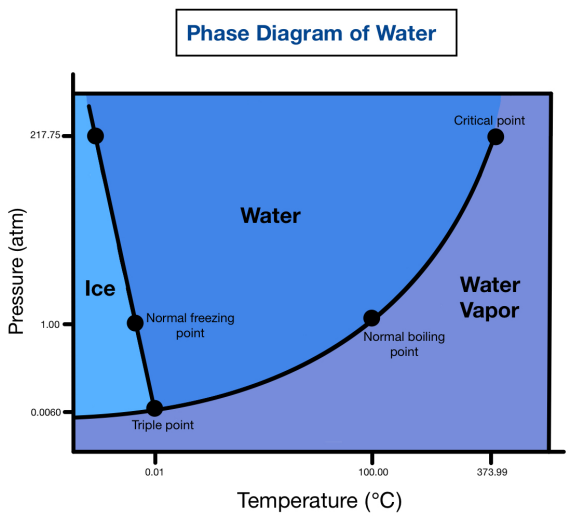

Figure 8: water phases

Now lets look at the change in K values for a variety of solutes in water across a range of temperatures:

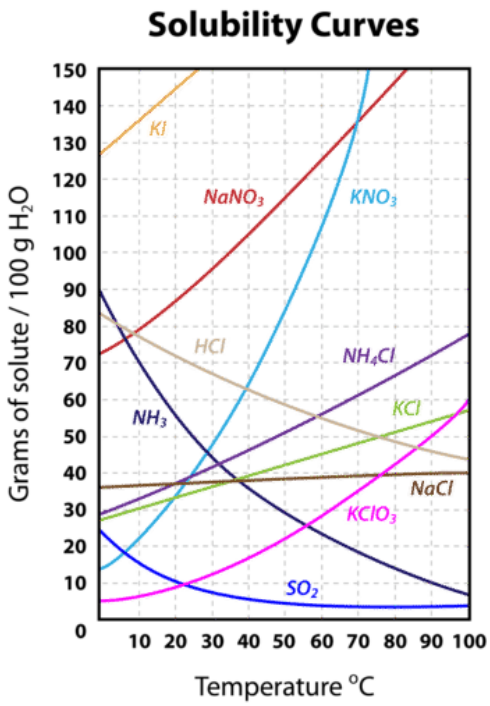

Figure 9: s curve

As you can see, the K_{sp} values of solid solutes tend to increase with increasing T because an addition of heat increases molecular motion and thus increases the statistical probability that the solid solute will enter the aqueous state.

In the context of gas solubility in a closed container, the K_H values of gases tend to decrease with increasing T. The outgassing process of gases requires the input of heat becauase gaseous molecules have higher KE than the same molecules in an aqueous state. With increasing T, this relative favoring of the gaseous state decreases $K_H = \frac{C}{P}$ by increasing the partial pressure (P) at the expense of the concentration (C).

Definition 5.1: Example

You may think that the increase in pressure also increases solubility, thus there must be some magnitude of a favoring of increased C. However, the shift toward the gas phase is so dominant that it overrides the fact that higher pressure usually helps solubility.